

Co-generating visions: inclusive representation and reflective visuality in the age of extended intelligences

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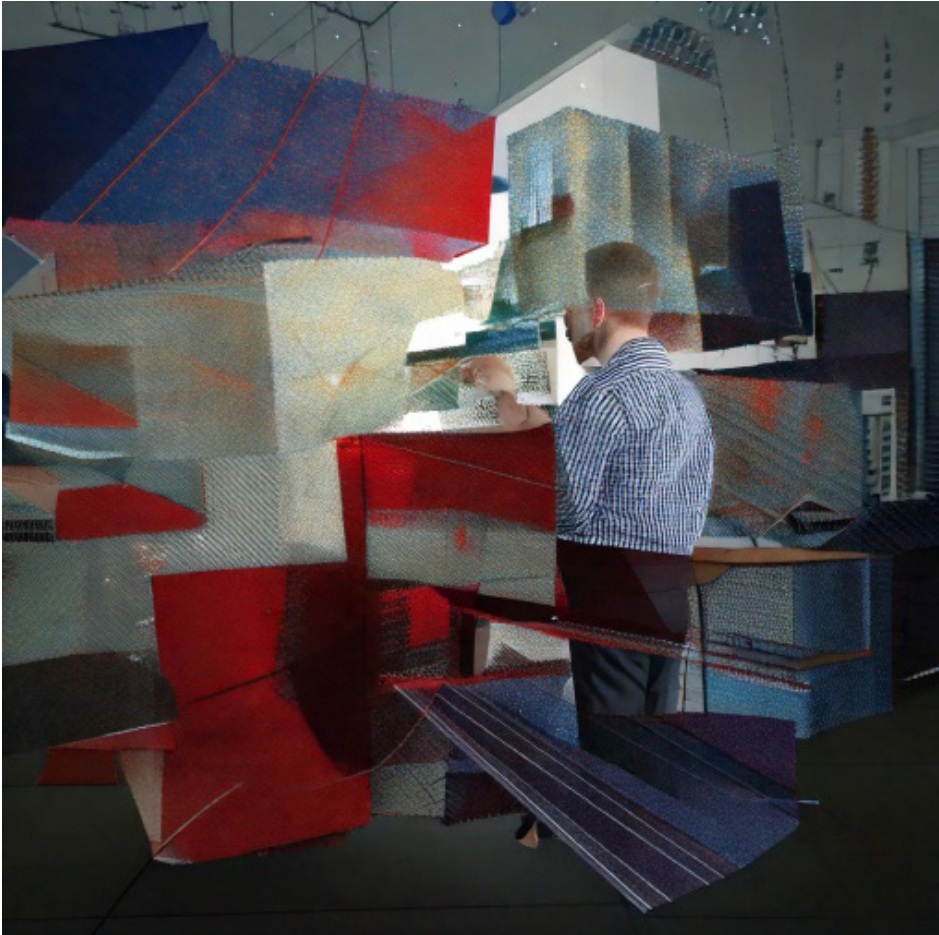
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ABSTRACT The integration of artificial intelligence into representational processes marks the transition from a 'sensitive' tool to a hybrid cognitive ecosystem, in which the relationship between human and machine generates new forms of situated knowledge. Within this framework, representation can become an act of perceptual and reflective co-generation, capable of interweaving attention, empathy, and care toward both the visible and the invisible dimensions of reality. This reflection emerges from the convergence of two research perspectives. On the one hand, reflective visuality is approached as a cognitive practice in which AI assumes the role of an epistemic counterpart: the collaboration between intelligences produces a process of co-observation that interrogates the limits of human vision and amplifies its interpretative capacity. Artificial intelligence introduces a computational aesthetic that mirrors the biases and cultural conventions embedded in datasets, acting as a critical lens capable of revealing visual tensions and contributing to the definition of new perceptual and morphological regimes. On the other hand, landscape representation is understood as a practice of inclusion and accessibility, capable of translating into visual form the sensitive relationships among body, mind, and space. Drawing and digital visualization become narrative tools that do not merely describe phenomenal reality but instead reveal its latent and perceptual dimensions. In this perspective, the embodied nature of experience and the ethical responsibility of care orient representation toward a political and collective act, in which seeing becomes a form of shared attention involving 'multiple intelligences': human (designer and users) and non-human (generative AI). AI thus emerges as a reflective and co-imaginative partner: a cognitive interlocutor that reframes representation as a critical and design-oriented space, where technology and human sensibility cooperate to generate new forms of perception, knowledge, and responsibility.



Introduction. Extended Intelligence as a Visual Ecosystem

The pervasive integration of artificial intelligence into representation processes marks a paradigmatic shift: from a technical tool to a hybrid cognitive ecosystem. It is no longer a question of using a device to translate or optimize a pre-existing idea, but of operating within an environment in which human and computational capabilities co-evolve, redefining the ways in which we see, interpret and design the world. In this perspective, AI is not a simple operational accelerator but an epistemic actor that intervenes in the construction of meaning, contributing to the formation of new visual and decision-making grammars.

The paradigm of extended intelligences invites us to overcome the dichotomy between author and tool, between subject and technology, to recognize representation as a relational space. The act of representing becomes a process of negotiation between different forms of

intelligence: the situated and bodily intelligence of the human and the algorithmic, statistical, predictive intelligence of the machine. In this condition, the gaze is no longer just a restitution of the visible but a co-production of scenarios, hypotheses and imaginaries.

It is in this horizon that the two research perspectives intertwined here are placed. On the one hand, reflective visibility investigates the contemporary empathic and perceptual crisis, questioning the role of artificial intelligence as a critical counterpart of the human gaze. Collaboration with generative systems becomes an opportunity to observe the limits, biases and conventions that structure our way of seeing, opening up the possibility of a co-imagination capable of articulating plural and conscious visions. On the other hand, the representation of the landscape is understood as a practice of inclusion and enhancement, oriented towards the care of the relationships between mind, body and space. In a context marked by the hybridization between the analog and

Experimental visual study on co-generated representation. The image returns a spatial configuration built through the superimposition of heterogeneous and partially misaligned logics, making explicit processes of selection, adaptation and renegotiation of the visible. AI-generated image (Adobe Firefly).

digital dimensions, artificial intelligence acts as a perceptual and narrative amplification device, making it possible to integrate quantitative data and experiential qualities, and to make visible thresholds, fragility and often marginalized potential. The aim of this paper is to outline a vision of representation as an act of cognitive and design co-generation. In this vision, humans and AI collaborate not only to produce forms, but to imagine ways of shared attention and collective responsibility. Representation thus becomes a care device: a critical space in which extended intelligence can contribute to building more empathetic, inclusive and conscious practices, both in the immaterial and in the material domain.

Visual Intelligences and Reflective Visibility

The transformation introduced by artificial intelligence in the processes of representation concerns first and foremost the regime of the gaze. As numerous

studies of visual culture state, each era is defined by a specific configuration of the observer, determined by technical devices, cultural infrastructures and epistemic models that structure what can be seen and how it can be seen. AI is part of this genealogy as a new optical-cognitive apparatus: not a simple prosthesis of the human gaze but a system capable of reorganizing the conditions of visibility itself. In contemporary visual culture, in fact, the image is not a stable object but a dynamic process within socio-technical networks. Following W.J.T. Mitchell (2018), images do not limit themselves to representing the world: they act, produce effects, organize relationships. With generative systems, this agency becomes radicalized. The image emerges as a probabilistic synthesis derived from massive datasets, statistical correlations and predictive models. The computational aesthetics described by Lev Manovich (2020) highlights how such systems produce emergent morphologies and new visual grammars, redefin-

Fig. 1 - Series of visual studies generated to test, in simulation form, co-designed representation processes. The images relate different logics of spatial organization, making observable phenomena of selection, interference and restructuring of the visual field. The sequence constitutes an analytical device for reading the conditions of visible production. AI-generated images (Adobe Firefly).



ing the parameters of contemporary design and formal culture.

In this scenario, reflective visibility coincides with the critical exploration of the conditions of seeing in the ecosystem of extended intelligences. Returning to the question of the situated vision (Haraway, 1988): there is no neutral gaze but partial and embodied positioning. With the integration of AI, this situation expands to include computational infrastructures, digital archives, and algorithmic logic. The representative act thus becomes distributed, the result of a co-construction between embodied perception and automated synthesis.

Therefore, the collaboration between human and machine can be understood as a practice of co-vision. AI does not 'see' in the phenomenological sense of the term, but analyses patterns, identifies recurrences, generates variations. However, in the dialogue with the designer, the algorithmic output operates as an epistemic counterpart: it returns a re-worked version of the world that makes visible cultural assumptions, hierarchies and biases inscribed in the data. In this sense, AI acts as a critical lens capable of exposing the invisible infrastructures of the contemporary imagination.

Co-generation, in this perspective, is not primarily about the multiplication of formal solutions but about the production of awareness about the process of seeing itself. Interacting with a generative system means dealing with a device that returns visual configurations derived from archives, datasets, correlations: each image is the result of a cultural sedimentation translated into probability. In this sense, co-generation can be understood as a form of thinking through images. It is not a matter of quickly obtaining a visualization but of using the generated image as a speculative device: a tool to observe deviations, recurrences, stereotypes, absences. AI, far from being an operational assistant, is configured as a speculative subject that participates in the redefinition of contemporary visual regimes.

In the design field, for example, the use of text-to-image systems to explore alternative visual scenarios often makes implicit structures of the gaze evident: prompts that describe social or spatial situations produce images that reiterate

cultural hierarchies, stereotyped representations of bodies or standardized urban configurations. In these cases, the output does not function as a formal solution but as a critical surface, capable of making visible the underlying logics of the datasets and activating a reflection situated on the conditions of seeing.

AI, far from being an operational assistant, is thus configured as a speculative subject that participates in the redefinition of contemporary visual regimes.

However, this dynamic also highlights a structural criticality: the algorithmic aestheticization of the visual. The reiteration of dominant patterns, inscribed in datasets and optimization logics, can consolidate homogeneous imagery and reduce complexity to statistically prevalent configurations. The risk is not only stylistic but epistemic: what is shown tends to coincide with what is more frequent, more trained, more computationally recognizable. This is where reflective visibility takes on a strategic function. Thinking about the image in the ecosystem of extended intelligences means governing (and not undergoing) the process of cognitive extension. The role of visual design is not to passively adhere to the predictive logic of models, but to introduce deviations, misalignments, questions. In other words, exercising a critical responsibility towards the conditions that make viewing possible.

This responsibility can be read in terms of care (Puig de la Bellacasa, 2017), thus implying attention to the relationships and systems of which we are part. Applied to human-machine co-vision, it translates into a vigilance on the infrastructures of the visible: datasets, parameters, algorithmic architectures. In this perspective, reflective visibility becomes a practice of maintenance of the visible: not accumulation of images, but conscious construction of perceptual regimes. AI is not only a creative partner but an interlocutor that forces the project to make its conditions of possibility explicit. The human-machine co-vision thus opens up a space in which aesthetics, epistemology and ethics intertwine, preparing the ground for a conception of representation as an act of situated attention and shared responsibility, both in the domain of images and in that of landscape and territorial relations.

Fig. 2 - Series of visual studies generated to test, in simulation form, co-designed representation processes. The images relate different logics of spatial organization, making observable phenomena of selection, interference and restructuring of the visual field. The sequence constitutes an analytical device for reading the conditions of visible production. AI-generated images (Adobe Firefly).



Care and inclusion of the landscape as representative practices

Representing a landscape means operating an act of care, aimed at giving voice to what remains marginalized: thresholds and experiences. This need for inclusion is not only an ethical instance, but a recognition of the diversity of cultural identities that interact with heritage. In this sense, we must consider the numerous person-context interactions, influenced by many factors, among which it is essential to mention: learning opportunities and concrete spaces with which contact takes place (Hayoe, 2019).

In this perspective, representation must take charge of the barriers that lead to passive exclusion, i.e. that form of marginalization that derives from structural deficiencies rather than from a deliberate intent. Unveiling the discontinuities of the landscape means, therefore, contrasting the identity of the different, which leads to an inability of the subjects to actively participate in life in the territory. The role of the accessible spatial configuration therefore

becomes fundamental in order to make direct experience inclusive, admitting the limit in not being able to satisfy every individual and at the same time identifying multisensory systems that can improve perceptual quality (Pallasmaa, 2012).

The integration of artificial intelligence marks a paradigm shift towards hybrid cognitive ecosystems. AI is not a simple tool, but an amplification device in an era like today's, which could be defined as onlife, where the digital and analog worlds merge in a hybrid environment (Floridi, 2023). This state requires a "re-ontologization" of reality, in which it is necessary to interpret space and relationships through new schemes; The information that allows the manipulation of the hybrid environment is contained in databases that deep learning is able to process in times and ways that greatly simplify the architect's work, leaving room for pure creativity.

AI makes it possible to reread the territory through sensory and narrative parameters. For example, practices that integrate street view imagery, geolocated data, and



Fig. 3 - Series of visual studies generated to test, in simulation form, co-designed representation processes. The images relate different logics of spatial organization, making observable phenomena of selection, interference and restructuring of the visual field. The sequence constitutes an analytical device for reading the conditions of visible production. AI-generated images (Adobe Firefly).

machine learning models make it possible to build maps of perception in which dimensions such as safety, accessibility, or a sense of belonging are displayed alongside physical data (Gong et al., 2025). These hybrid representations make critical issues that are difficult to detect through traditional tools, such as perceptual barriers or experiential discontinuities, legible, offering the project a more inclusive and situated knowledge base. This approach makes it possible to go beyond the purely technical vision to incorporate qualitative as well as quantitative elements, coming to redefine the lived space through the real perception of it.

Drawing with AI means cogenerating situated knowledge, in a dialogue between man and machine, where the architect assumes a role based on the real needs of users. The practice of integrated drawing becomes a way to slow down the gaze, allowing us to observe and access a deeper understanding of the territory as a living system. This collaboration does not erase authorship but expands it through digital storytelling. There are numerous theories

that explain the concept of coherence between memory and identity, and integrated storytelling guarantees the user to be able to process images that would otherwise risk remaining buried in memory, thus creating one or more stories, through tools of meaning and explanation that structure our identity (Lambert, 2010). Using AI becomes essential to configure the landscape, building digital Memory Boxes, where multimedia elements help to reconstruct complete memories and expand the narrative repertoire of the project.

The goal must be to support sustainable development, understood as an integrated process aimed at:

- Improve the perceptual quality of space and in this sense, AI can act as a powerful enabler;
- Systematize the process of managing places, ensuring safe and controlled spaces (Vinuesa et al., 2020);
- Provide processes based on principles of transparency and ethical standards to avoid exacerbating inequalities or reproducing social biases inherent in datasets.

Representative practice is reconfigured, orienting itself towards the semantic negotiation of what defines the sustainability and moral correctness of the future human habitat (Floridi, 2023). In this perspective, inclusive representation, enhanced by systems of extended intelligence, transcends the mere mimetic or phenomenological restitution of the visible to become a language of empathic mediation. It implements a methodology that we could define as “circular listening” (Lambert, 2010, pp. 18), capable of extracting and decoding the latent narratives that reside in the frictions between the material datum and its digital abstraction. The convergence between the analytical rigor of machine learning and the situated dimension of body perception thus makes it possible to structure a grammar of attention: a system of signs capable of preserving the stratified complexity of the landscape and mapping the density of the human relationships that substantiate it, transforming drawing into an act of responsibility towards the landscape.

Applications developed in the academic and institutional fields show how AI can support a representation of the landscape oriented towards care and inclusion, making visible perceptual and relational dimensions often excluded by traditional tools. Systems applied in real cities such as London (United Kingdom), Xi'an (China), Nice (France) and Vienna (Austria) translate subjective judgments of safety, accessibility and livability into operational spatial representations, configuring a co-vision between human evaluations and algorithmic synthesis. Similarly, landscape values mapping projects applied to real territories integrate visual and narrative data to produce maps of affective and symbolic value of the landscape, while experiments in the field of built heritage show how generative AI can articulate polyvocal narratives of places, transforming representation into an act of attention and responsibility towards territorial and memorial complexity.

Conclusions: Towards a responsible co-vision

The paper proposed to read the integration of artificial intelligence in the processes of representation as a transition from a tool to a hybrid cognitive ecosystem, in which

the production of the visible is no longer attributable to a single subject but emerges from the interaction between embodied perception, computational infrastructures and material contexts. In this framework, representation is configured as a practice of shared attention: a relational device in which human, machine and environment co-produce conditions of visibility, interpretation and decision.

The first chapter highlighted how AI acts on the gaze regime, reorganizing the conditions of visibility and making the image a dynamic process inscribed in socio-technical networks. The notion of co-vision has made it possible to understand algorithmic output not as a simple execution but as an epistemic counterpart: a critical surface that puts archives, datasets and cultural assumptions in tension, making biases, recurrences and absences perceptible. In this perspective, co-generation becomes a thought in images: a speculative and reflective use of visual generation.

The second chapter shifted the focus to representation as a practice of care and inclusion. Here, artificial intelligence has been taken as a perceptual and narrative amplification device in an onlife environment, capable of connecting physical data and qualitative dimensions of experience. Inclusive representation, supported by intelligent tools, does not coincide with the mimetic restitution of space but with the construction of a grammar of attention that makes thresholds, vulnerabilities, memories and possibilities of access visible. In this sense, design, integrated by computational infrastructures, is configured as a design act that explicitly assumes responsibility for the relationships between bodies, places and systems.

From this convergence emerges a common thesis: AI is an opportunity to rethink both the visual project as forms of collective responsibility. It is not a question of entrusting the production of the image or the management of space to calculation, but of consciously assuming extended intelligence as a contemporary condition of representation. What is at stake is not only methodological but ethical: it concerns what is made visible, what is excluded and the ways in which representation can guide fairer, more transparent and shared practices.

The final vision proposed is therefore this: co-generating situated visions

means inhabiting a common horizon in which representation becomes a space of negotiation between human, machine and environment. A horizon that requires visual awareness, critical capacity and design that is attentive to differences, so that artificial intelligence does not consolidate stereotypes and marginality but contributes to building conditions of empathy, accessibility and responsibility. In this framework, representation is not only an outcome but a practice of care: a

way of seeing together that makes it possible to imagine (and design) more habitable forms of the world.

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